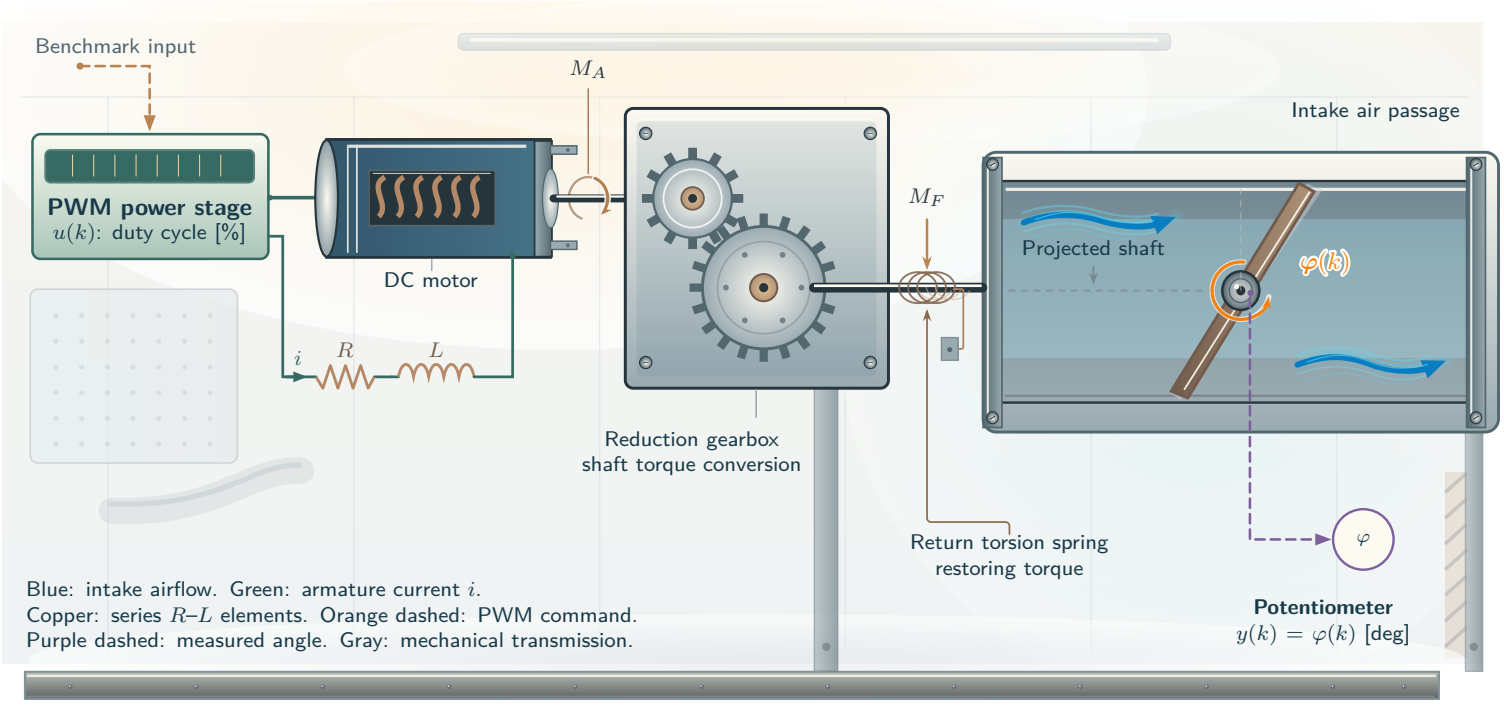
# Automotive throttle: actuation and identification

Paper section 3.4: *Electro-mechanical throttle*

## Physical chain: PWM drive, DC motor, gear box, spring and throttle plate




Automotive engine-test-cell rig (illustrative equipment and lighting; not to scale)

**Butterfly plate**
variable opening

**Acquisition**
Sample interval: $T_0 = 10$ ms.

Input: PWM duty [%].
Output: angle [deg].

Published plate range:
$10^\circ$ closed to $90^\circ$ open.

**Identification challenge**
Friction, nonlinear spring and mechanical compliance can change the input–output response.

No calibrated internal parameters are inferred here.

**Training experiment**
Multisine excitation; $N = 10\,000$.
Fit dynamic models using the training record only. Keep measured duty and angle in their stated units.

**Independent validation**
Quasi-APRBS excitation; $N = 2\,500$.
Evaluate one-step and free-run prediction separately; report errors, residuals and model complexity.

**Repository evidence**
PWA/TS-inspired educational routines are available. Original measured time series have not been recovered; fixtures are synthetic.

Technology illustration, not a calibrated throttle model or a production assembly drawing. Gear ratio, motor constants, spring law and friction are not inferred. See `THROTTLE_RECOVERY.md` for data access status.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.